10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 6 — Reproductive System

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. In which part of the body oogenesis takes place?

(a) Oviduct (b) Ovary
(c) Uterus (d) Fallopian tube

Answer: (b) — The formation of the female gametes (eggs or ova) is called oogenesis, and it takes place in the ovary inside a small, fluid-filled sac called the follicle. The fallopian tube only transports the egg, and fertilization takes place there.

2. Which of the following is NOT a part of the male reproductive system?

(a) Testes (b) Uterus
(c) Epididymis (d) Prostate gland

Answer: (b) — The uterus is a hollow, muscular organ of the female reproductive system in which the fertilized egg implants and develops into a foetus. Testes, epididymis and prostate gland are all parts of the male reproductive system.

3. Which hormone directly triggers ovulation?

(a) Oestrogen (b) Progesterone
(c) Luteinizing hormone (LH) (d) Follicle-stimulating hormone (FSH)

Answer: (c) — In females the luteinizing hormone triggers ovulation, that is, the release of an egg from the ovary, and it also stimulates the production of progesterone. FSH stimulates the growth of the follicles.

4. Which hormone is responsible for the development of male secondary sexual characteristics?

(a) Follicle Stimulating Hormone (b) Oestrogen
(c) Testosterone (d) Luteinizing Hormone

Answer: (c) — Testosterone is produced by the testes and it initiates and regulates the development of the male secondary sexual characteristics during puberty, such as the deepening of the voice, the growth of facial and body hair and an increased muscle mass.

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5. What is the function of acrosome?

(a) Guides the sperm towards egg cell
(b) Releases enzymes that help sperm penetrate the egg's outer layers
(c) Activates the egg for fertilization
(d) Helps in the fusion of sperm and egg nuclei

Answer: (b) — The acrosome is a cap-like structure on the head of the sperm. It releases enzymes which digest the outer layers of the egg, so that the sperm can reach the egg cell membrane and fertilize it.

6. What is the function of the Fallopian tubes?

(a) Site of fertilization
(b) Storage of eggs
(c) Secretion of oestrogen
(d) Transport of sperm to testes

Answer: (a) — The two fallopian tubes transport the egg from the ovary to the uterus, and fertilization takes place inside the fallopian tube when a sperm fuses with the egg. Eggs are produced in the ovaries and oestrogen is secreted by the ovaries.

7. Which hormone is primarily responsible for the development of male secondary sexual characteristics?

(a) Oestrogen
(b) Testosterone
(c) Progesterone
(d) FSH

Answer: (b) — Testosterone, secreted by the testes, is the hormone which develops the male secondary sexual characteristics such as the deep voice, beard and body hair. It is also essential for spermatogenesis.

B. Write short answers.

1. Complete the chart of spermatogenesis by mentioning the names of cells and number of chromosomes (2n or 1n).

The chart of spermatogenesis, from the first cell to the sperm, is completed as follows:

Stage in the chart	Name of cell	Chromosomes
First cell (gamete mother cell)	Primary spermatocyte	2n (diploid)
The two cells formed after meiosis-I	Secondary spermatocytes	1n (haploid)
The four cells formed after meiosis-II	Spermatids (immature sperms)	1n (haploid)
The four cells after maturation	Sperms	1n (haploid)

Spermatogenesis takes place in the seminiferous tubules of the testes, and it is stimulated by follicle-stimulating hormone (FSH) along with testosterone.

2. Complete the chart of oogenesis by mentioning the names of cells and number of chromosomes (2n or 1n).

The chart of oogenesis, from the first cell to the egg, is completed as follows:

Stage in the chart	Name of cell	Chromosomes
First cell (gamete mother cell) in the follicle	Primary oocyte	2n (diploid)
The larger cell formed after meiosis-I	Secondary oocyte	1n (haploid)
The smaller cell formed after meiosis-I	First polar body	1n (haploid)
The cell formed after meiosis-II	Egg (ovum)	1n (haploid)
The small cell which degenerates after meiosis-II	Second polar body	1n (haploid)

Oogenesis takes place in the ovary inside the follicle, and it is stimulated by the follicle-stimulating hormone released by the anterior lobe of the pituitary gland.

3. Name the parts of the testis and ovary where gamete formation takes place.

In the **testis**, the male gametes (sperms) are formed in the **seminiferous tubules**. In the **ovary**, the female gamete (egg or ovum) is formed inside a small, fluid-filled sac called the **follicle**.

4. What happens to the egg after it is released from the ovary?

After ovulation the egg is released from the ovary and is picked up by the fallopian tube. The fallopian tube transports it towards the uterus, and fertilization takes place inside the fallopian tube if a sperm reaches the egg. The sperm releases enzymes from its acrosome to penetrate the outer layers of the egg, the nuclei of the sperm and the egg fuse, and a diploid zygote is formed. This zygote then moves to the uterus, where it implants in the wall of the uterus and develops into a foetus. If the egg is not fertilized, it breaks down and passes out of the body with the blood and tissue of the uterus wall.

5. Compare spermatogenesis and oogenesis in terms of the number of gametes produced.

In **spermatogenesis**, one primary spermatocyte divides by meiosis-I to form two secondary spermatocytes, and each of them divides by meiosis-II. So **four haploid sperms** are produced from one primary spermatocyte, and all of them mature into motile sperms. In **oogenesis**, one primary oocyte divides by meiosis-I into one large secondary oocyte and one small first polar body, and after meiosis-II the secondary oocyte gives one egg and one second polar body. Thus **only one functional egg** is produced from one primary oocyte, and the polar bodies degenerate. So spermatogenesis gives four gametes per mother cell, while oogenesis gives only one gamete per mother cell.

6. Describe two major ways through which HIV can be transmitted.

(i) Through sexual contact: HIV spreads mainly through unprotected sexual contact with an infected person, because the virus is present in the body fluids.

(ii) Through infected blood or needles: the virus also spreads by sharing contaminated needles, for example among drug users, and by the transfusion of infected blood. In Pakistan, unsafe injections and unregulated blood transfusions are important reasons for the increasing number of HIV cases.

It can also pass from an infected mother to her child during childbirth or breastfeeding.

7. Differentiate between: (i) Oestrogen and progesterone

Oestrogen	Progesterone
It is primarily produced by the ovaries before ovulation.	It is produced by the ovaries after ovulation.
It initiates and regulates the development of the female secondary sexual characteristics during puberty, such as breast development and widening of the hips.	It prepares the walls of the uterus for the implantation of the fertilized egg.
It helps in the onset of the menstrual cycles and regulates the female reproductive cycle.	It maintains pregnancy by preventing contractions of the uterus.

(ii) Spermatogenesis and oogenesis

Spermatogenesis	Oogenesis
The formation of male gametes (sperms).	The formation of the female gamete (egg or ovum).
It takes place in the seminiferous tubules of the testes.	It takes place in the ovary, inside the follicle.
Four functional gametes are produced from one mother cell.	Only one functional gamete is produced from one mother cell; the polar bodies degenerate.
It starts at puberty and continues throughout life.	The formation of ova begins before birth and stops at menopause.
The gametes formed are motile sperms.	The gamete formed is a large, non-motile egg.

(iii) Spermatocyte and oocyte

Spermatocyte	Oocyte
It is the gamete mother cell present in the testes.	It is the gamete mother cell present in the ovary.
Its division produces sperms.	Its division produces the egg (and polar bodies).
By meiosis, one primary spermatocyte gives four sperms.	By meiosis, one primary oocyte gives one egg and polar bodies.
It is small in size, because no food store is needed for the sperm.	It is much larger in size, because the egg stores food for the early embryo.

C. Write answers in detail.

1. Describe the role of hormones in male sexual development.

The male sexual development is controlled by four hormones.

(i) Testosterone: it is primarily produced by the testes. It initiates and regulates the development of the male secondary sexual characteristics during puberty, such as the deepening of the voice, the growth of facial and body hair and an increased muscle mass. It is also essential for spermatogenesis.

(ii) Follicle-stimulating hormone (FSH): it is produced by the anterior pituitary gland, and it stimulates the testes to produce sperms.

(iii) Luteinizing hormone (LH): it is also produced by the anterior pituitary gland, and it stimulates the testes to produce testosterone.

(iv) Gonadotropin-releasing hormone (GnRH): it is released by the hypothalamus, and it stimulates the pituitary gland to release LH and FSH. In this way the hypothalamus controls the whole process of male sexual development and sperm production.

2. Describe the role of hormones in female sexual development.

(i) Oestrogen: it is primarily produced by the ovaries. It initiates and regulates the development of the female secondary sexual characteristics during puberty, such as breast development, widening of the hips and the onset of the menstrual cycles, and it also regulates the female reproductive cycle.

(ii) Progesterone: it is produced by the ovaries after ovulation. It prepares the walls of the uterus for the implantation of the fertilized egg and maintains pregnancy by preventing contractions of the uterus.

(iii) Follicle-stimulating hormone (FSH): it is released from the anterior pituitary gland, and in females it stimulates the growth of the follicles, which contain the eggs, and also stimulates the production of oestrogen.

(iv) Luteinizing hormone (LH): it triggers ovulation, that is, the release of the egg from the ovary, and it stimulates the production of progesterone.

(v) Gonadotropin-releasing hormone (GnRH): it is released by the hypothalamus and stimulates the pituitary gland to release LH and FSH.

3. Describe the process of spermatogenesis.

The formation of the male gametes (sperms) is called spermatogenesis, and it takes place in the seminiferous tubules of the testes. Follicle-stimulating hormone along with testosterone stimulates the production of sperms; FSH is produced by the anterior pituitary gland and testosterone by the testes.

In the seminiferous tubules, the diploid gamete mother cells called primary spermatocytes are present. Each primary spermatocyte divides by meiosis-I and forms two haploid secondary spermatocytes. Each secondary spermatocyte then divides by meiosis-II, and in this way four haploid spermatids are formed. These spermatids are immature sperms; changes occur in them and they mature into motile sperms. In this way one primary spermatocyte gives four functional sperms, each with a haploid number of chromosomes.

4. Describe the functions of the major parts of male reproductive system.

Part	Function
Testes	A pair of oval-shaped glands which produce the sperms and the male hormone testosterone.
Epididymis	A long, coiled tube attached to each testis, where the sperms are stored and matured.
Vas deferens	A muscular tube which transports the mature sperms from the epididymis to the urethra.
Seminal vesicles	Two small glands behind the bladder which produce a sugary fluid to nourish the sperms.
Prostate gland	A walnut-sized gland below the bladder which adds protective fluid to the semen.
Urethra	A tube inside the penis which carries the semen and the urine to the outside of the body.
Penis	An external organ which transfers the sperms into the female reproductive tract.

5. Describe the functions of the parts of female reproductive system.

Part	Function
Ovaries	A pair of almond-shaped glands which produce the eggs (ova) and the hormones oestrogen and progesterone.
Fallopian tubes	Two tubes which transport the eggs from the ovaries to the uterus; fertilization takes place here.
Uterus	A hollow, muscular organ where the fertilized egg implants and develops into a foetus during pregnancy.
Cervix	The lower, narrow part of the uterus.
Vagina	A muscular canal which connects the cervix to the outside.

6. Explain AIDS as an example of sexually transmitted diseases.

Sexually transmitted diseases (STDs) are the infections which are mostly transmitted through sexual contact, and they can be caused by bacteria, viruses or parasites. Common STDs are HIV/AIDS, syphilis and herpes. Many STDs have no symptoms, so a person may carry and spread the infection without knowing it, and if they are left untreated they can cause infertility and organ damage.

AIDS is caused by the Human Immunodeficiency Virus (HIV). HIV attacks and weakens the immune system of the body, so the patient cannot fight infections and certain cancers. HIV spreads mainly through sexual contact, sharing of contaminated needles, transfusion of infected blood, and from an infected mother to her child during childbirth or breastfeeding. In Pakistan, HIV cases are increasing particularly because of unsafe injections, unregulated blood transfusions and a lack of awareness among young people.

The early signs and symptoms are fever, fatigue, swollen lymph nodes and skin rashes. As the disease progresses the patient becomes vulnerable to severe infections, weight loss, chronic diarrhoea, pneumonia and certain cancers. There is no cure for AIDS, but antiretroviral therapy (ART) can control the virus, strengthen the immune system and help the patient live a longer and healthier life.

D. Inquisitive questions

1. Why only one egg is usually produced from one primary oocyte?

In oogenesis the division of the primary oocyte is unequal. During meiosis-I it divides into one large secondary oocyte and one small first polar body, and during meiosis-II the secondary oocyte again divides unequally into one large egg and one small second polar body. The two polar bodies contain very little cytoplasm and they degenerate. This happens because the egg has to store a large amount of food and cytoplasm for the early development of the embryo, and this food cannot be divided equally among four cells. Therefore only one functional egg is produced from one primary oocyte.

2. How does the structure of a sperm cell relate to its function?

A sperm is a very small, motile cell with a haploid nucleus, and every part of it helps it to reach and fertilize the egg. Its head contains the nucleus, which carries the paternal chromosomes, and the acrosome on the head releases enzymes that digest the outer layers of the egg so that the sperm can enter it. Behind the head there are many mitochondria, which provide the energy needed for movement, and a long tail which beats and pushes the sperm forward through the female reproductive tract. Its small size and streamlined shape reduce resistance and help it to swim quickly. Thus the structure of the sperm is perfectly suited to its function of carrying the male genetic material to the egg.

3. What steps can individuals take to prevent HIV transmission?

HIV spreads through body fluids, so its transmission can be prevented by avoiding these routes. A person should avoid unprotected sexual contact and should be faithful to one partner. Needles and syringes should never be shared; only disposable, sterile needles should be used, especially for injections and for drug use. Blood should be transfused only after it has been screened for HIV. Pregnant women should be tested for HIV, because treatment of the mother greatly reduces the risk of passing the virus to the child. Awareness and education about HIV, and the use of sterile equipment by barbers, dentists and health workers, are also very important.

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